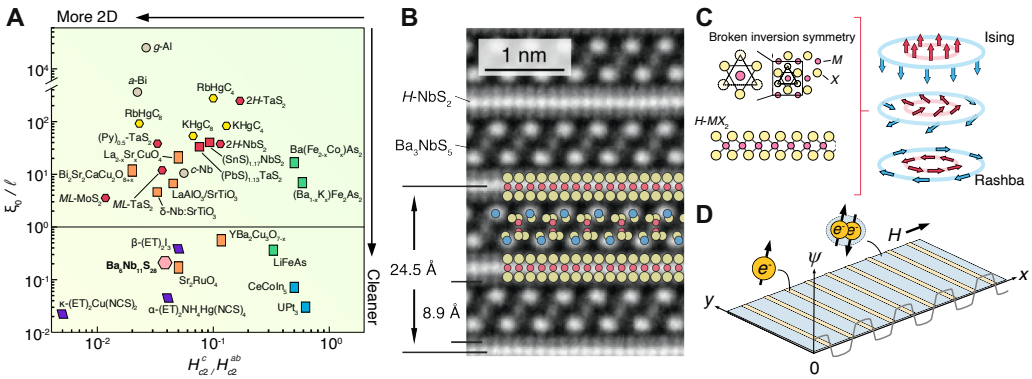

Figure 1: (a) Survey of superconducting materials sorted by cleanliness and two-dimensionality. (b) Natural superlattice of H-NbS_2 and Ba_3NbS_5. (c) Broken inversion symmetry in H-NbS_2 layer and broken mirror from adjacent Ba_3NbS_5 layers leads to mixed Rashba and Ising spin-orbit coupling. (d) An FFLO state characterized by a real-space modulation of the order parameter ψ could appear with in-plane magnetic field H.

Scientific Importance

Dimensionality and disorder have long been of interest in the study of superconductivity. Figure 1(A) presents a historical survey of superconducting materials sorted according to these parameters, the former characterized by H^c_{c2}/H^{ab}_{c2} (the ratio of the upper critical magnetic fields perpendicular and parallel to the 2D layer) and the latter by ξ_0/l (the ratio of the Pippard coherence length and the electronic mean free path). The sparsely populated regime of clean superconductivity is occupied by materials that exhibit some of the most exotic forms of superconductivity [1,2]. We have recently discovered a new superconductor $Ba_6Nb_{11}S_{28}$ that belongs to this rare class of materials [3]. As shown in Fig. 1(B), the material is composed of inversion symmetry breaking *H*-NbS_2 layers separated by insulating Ba_3NbS_5 layers. The insulating layers break the mirror symmetry of the NbS_2 layer plane, giving rise to a mixed Rashba and Ising spin-orbit texture (Fig. 1(C)). This mixture of spin-orbit coupling, clean-limit superconductivity, and two-dimensionality could host unconventional pairing [4]. In the presence of in-plane magnetic field, $Ba_6Nb_{11}S_{28}$ could yield the long-pursued Fulde-Ferrell-Larkin-Ovchinnikov (FFLO) state (characterized by a real-space modulation of the superconducting order parameter, see Fig. 1(D)) [5].

Studying the flux-line lattice (FLL) of this new, potentially unconventional superconductor is of great interest and could provide significant insight into the superconducting gap symmetry. It is well-established that the FLL can show deviations from the hexagonal Abrikosov lattice [6] in unconventional superconductors. SANS is a powerful technique to detect these deviations as demonstrated by studies of UPt_3, $YBa_2Cu_3O_{7-x}$, and $CeCoIn_5$ [7–9]. Analogously, the FFLO phase (depicted in Fig. 1(D)) may also be seen using SANS. The most heavily studied candidates, $CeCoIn_5$ and κ-$(ET)_2Cu(NCS)_2$, possess limitations that interfere with experimental probes of the FFLO state [10]. In particular, $CeCoIn_5$ has large local magnetic moments in the proposed FFLO regime and κ-$(ET)_2Cu(NCS)_2$ is physically fragile, limiting the applicable experimental probes. $Ba_6Nb_{11}S_{28}$ is a robust inorganic material that offers an exciting new platform to search for FFLO physics.

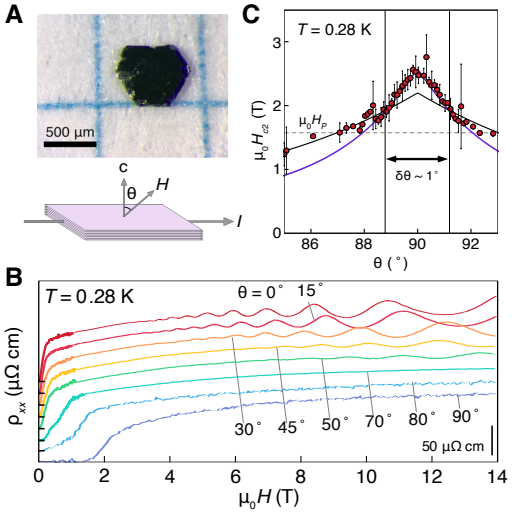

Figure 2: (A) top, Image of typical $Ba_6Nb_{11}S_{28}$ single crystal. bottom, angle θ is measured relative to crystallographic c-axis. (B) Resistivity as a function of field H measured at T = 0.28 K at various fixed θ. The superconductivity survives to higher fields for larger angles due to low-dimensionality of the system. (C) Upper-critical field μ_0H_{c2} close to θ = 90°. Within an angular window δθ ~ 1 °, the upper-critical field is enhanced possibly due to the appearance of a field-induced FFLO phase.

Both the FLL and FFLO patterns can be imaged by small-angle neutron scattering (SANS). At $T = 0.3$ K we estimate $q \sim 0.0015$ Å^{-1} and $q \sim 0.0025$ Å^{-1}, respectively; both are accessible at GP-SANS. This length scale and the need for ^{3}He temperatures with in-plane magnetic fields precludes scanning tunneling microscopy from probing either the FLL or FFLO state. $Ba_6Nb_{11}S_{28}$ and SANS could be a unique confluence of potentially exotic physics, an accessible material, and an ideal experimental technique.

Preliminary Work

Figure 2(A, top) shows a typical single crystal of $Ba_6Nb_{11}S_{28}$ with well-developed hexagonal habit measuring approximately 1×1×0.1 mm^3 in volume. Figure 2(B) shows longitudinal resistivity as a function of magnetic field, $\rho_{xx}(H)$, at $T = 0.28$ K measured for various fixed angles θ; the field angle θ is measured with respect to the crystallographic *c*-axis, Fig. 2(A, bottom). Superconductivity and Shubnikov-de Haas quantum oscillations of resistivity coexist in $Ba_6Nb_{11}S_{28}$, indicating clean-limit superconductivity. As discussed above, this combination can potentially support unconventional superconducting pairing.

When field is rotated towards the layer plane, θ = 90º, the zero-resistance state persists to higher fields, characteristic of reduced dimensionality. Figure 2(C) shows the upper-critical field μ_0H_{c2}, defined as the

field where ρ_{xx} reaches half of the normal state value, at $T = 0.28$ K for angles close to the layer plane. We find that the angular dependence does not fall onto a single fitting line (the cusp-like behavior at $\theta = 90^o$ leads us to use the 2D Tinkham model [11]) and instead requires two separate fits for angles outside (Fig. 2(C, black)) and inside (Fig. 2(C, purple)), the angular window $\delta\theta \sim 1^o$. The enhancement of $\mu_0 H_{c2}$ within $\delta\theta$ of 90º is reminiscent of behavior seen in the organic FFLO candidate α-$(ET)_2NH_4Hg(NCS)_4$ and predictions for the FFLO state in two-dimensional superconductors [10]. Aligning θ within this angular window is critical to observe the potential FFLO. Through preliminary SANS experiments at ORNL, we have identified a straight-forward approach to precisely align the sample which we describe below. Single crystal samples are immediately available for the proposed experiment and we are synthesizing and optimizing additional crystals.

Choice of Instrument

GP-SANS is the best instrument of choice for this proposal due to it being able to reach the lowest q of the SANS instruments at ORNL and support high field, low temperature sample environments. The sample environment also supports precise sample rotation, allowing us to tune the measurement geometry *in-situ* as we the necessary measurement geometry which we describe below.

Experiment Plan

We propose a SANS measurements in the 11 T horizontal cryomagnet with a ^{3}He refrigerator insert. This system allows for *in-situ* sample and field rotation, allowing us to probe the flux-line lattice and look for the potential FFLO behavior identified by transport above. We plan to load an aligned 5 x 5 mm^2 single crystal mosaic built using approximately 10 samples. Due to the well-developed hexagonal habit of the crystals, see Fig. 2(A, top), construction of the mosaic is not expected to be difficult. As stated above, during a preliminary SANS experiment we noted the need for precise orientation of the sample with respect to the field.

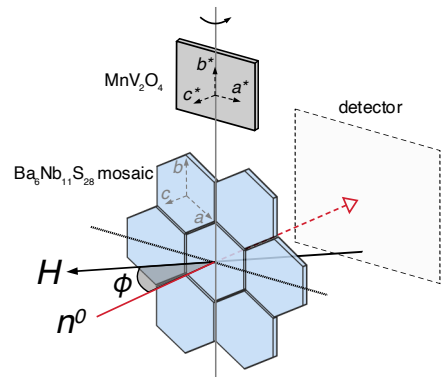

Figure 3: Measurement geometry for proposed SANS experiment. Using the MnV_2O_4 crystal, neutrons will be aligned along the crystal mosaic c-axis. The magnetic field can be rotated through the angle, φ, allowing us to probe both the flux-line lattice and the possible FFLO phase.

We improve upon this earlier experiment by co-mounting a single crystal of MnV_2O_4, see Fig. 3, to precisely align the sample plate with the magnetic field. MnV_2O_4 shows ferrimagnetic ordering with typical length scale ~ 100 nm that produces strong fin-like scattering at zero-field along the a^* and b^* axes [12]. The intensity of the fin-like scattering is maximized with neutrons along the c-axis of the MnV_2O_4 crystal (c^* in Fig. 3). The relative alignment of the MnV_2O_4 sample to the $Ba_6Nb_{11}S_{28}$ mosaic will be checked in-house with x-ray scattering prior to the main beamline experiment.

We will study scattering in two measurement geometries: with H along the c-axis and along the ab-plane of the mosaic, $\varphi = 0^o$ and $\varphi = 90^o$ in Fig. 3, respectively. As noted above, the ^{3}He refrigerator and 11 T horizontal magnet can continuously tune between both measurement geometries, providing great flexibility. We believe the proposed alignment procedure and this flexibility can yield H aligned within 0.5° of the mosaic. It may also permit pursuit of even finer features that are predicted to appear in the FFLO state within $\delta\theta$ [13]. We request 5 days of GP-SANS beamtime. During the first day, we aim to align the sample plate using the MnV_2O_4 crystal and begin mapping the temperature and field dependence of the FLL of $Ba_6Nb_{11}S_{28}$ (including control data in the normal state) in the $\varphi = 0^o$ geometry. We expect mapping the FLL will take approximately 3 days after which we will transition to the $\varphi = 90^o$ to search for the possible FFLO phase. Further experiments to map the evolution of the FFLO q with elevated temperatures and finite field tilt angle would be exciting to probe predictions for the FFLO phase and could connect with analogous states expected in high-energy matter [14].

Data Analysis and Scientific Outcomes

The observation of a flux-line lattice with symmetry deviating from the hexagonal Abrikosov state could indicate unconventional superconducting pairing. By fitting the 2D data in regions of interest near the expected q for the FLL, we can extract peak positions and identify the FLL symmetry. By comparing these results to theoretical calculations [6], we can infer the underlying pairing symmetry. In line with earlier experiments [7–9], the possible formation of flux lattice domains must be considered. With the ability to tune field, temperature, and angle, we can systematically rule out such extrinsic effects. In the $\varphi = 90^o$ geometry where we can search for the potential field-induced FFLO phase, the observation of any neutron scattering wave vectors upon application of sufficient in-plane magnetic field would be smoking gun evidence for the real space modulation associated with this phase, Fig. 1(D). Any such observations would be the first direct probe of the FFLO modulation, potentially validating a long-standing model for exotic superconductivity. More broadly, this system has the necessary ingredients to realize unconventional pairing and a field-induced FFLO state; SANS can potentially probe both of these scenarios. We believe the richness of this material and the power of SANS make this experiment extremely worthwhile.

Safety Considerations

For this experiment we will follow the subject matter expert on the ^{3}He refrigerator and cryomagnet. Along with any safety that is put in place with dealing with cryogens.

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